

CROP DISEASE PREDICTION DATA REPORT

AGRODATA PROJECT

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1. Business Understanding

1.1 Executive Summary

Agriculture remains the backbone of many economies, particularly in developing regions. However, pests and diseases significantly impact crop yields, often going undetected until it is too late. Farmers frequently rely on experience or guesswork, leading to misdiagnosis and excessive pesticide use. This not only reduces productivity but also harms the environment and increases costs.

Modern artificial intelligence (AI) technologies can bridge this gap by enabling early detection of plant diseases and pests to offer tailored treatment recommendations, minimize crop loss, ensure food security, and support sustainable farming practices. By leveraging computer vision and machine learning models trained on real-world agricultural datasets like TOM2024, we can help farmers make informed, timely decisions.

1.2 Problem Statement

Farmers often struggle to accurately identify crop diseases and pests, leading to delayed or incorrect treatments that reduce yield and increase costs. Therefore, there is a need for an AI-driven solution that can rapidly detect crop issues from images and provide actionable treatment recommendations.

1.3 Business Objectives

1.3.1 Main Objective

To develop an AI-powered system that accurately detects crop diseases and pests from images.

1.3.2 Specific Objectives

1. To develop a deep learning model to identify crop diseases and pests from images.
2. To preprocess image data to improve model generalization across various conditions.
3. To minimize the reliance on chemical pesticides by promoting early detection and precise intervention strategies.
4. To generate appropriate treatment or action recommendations for each predicted disease or pest, based on agricultural best practices.

2. Data Understanding

The project utilizes the 'TOM2024' dataset, a comprehensive and high-quality collection of crop images specifically curated for pest and disease detection in maize, tomato, and onion crops. It is a valuable resource for advancing agricultural research and AI-driven crop protection. It contains 25,844 raw images and 12,227 labeled images of tomato, onion, and maize crops, meticulously categorized into 30 distinct classes, representing various diseases, pests, and health conditions across the three crops.

The dataset used in this study is divided into two broad categories: category A and category B. Category A consists of image folders grouped by crop type and issue type (pests, diseases, or pest activities). It serves as the initial labeled dataset for

exploration, augmentation, or additional curation. The folders have subfolders containing images representing a specific pest, disease, or activity affecting the respective crop. These images are labeled and may vary in lighting, background, and clarity to simulate real-world field conditions.

On the other hand, category B is structured to support supervised machine learning and deep learning model training. Each crop folder contains train and test subfolders, and within those are class folders labeled according to the condition:

_d → Disease

_p → Pest

_a → Pest Activity (maize only)

A separate folder for healthy plants is included in each case.

Our focus has been on CATEGORY_B, which includes three separate datasets: maize_dataset, onion_dataset, and tomato_dataset. Each dataset is further split into train and test folders, and within each, images are organized by class labels corresponding to specific plant conditions or diseases. For example, the tomato dataset includes classes like blossom_end_rot_d, tomato_late_blight_d, mite_d, and healthy_leaf.

Initial inspection revealed inconsistencies in class folder naming (e.g., helthy_leaf vs healthy_leaf, mite_d vs Mite_d). These inconsistencies were resolved by merging and renaming folders to ensure uniformity across the training and test data. After cleaning, class distribution was reviewed to ensure a reasonable balance, although minor class imbalances were observed.

3. Data Preparation

Images were standardized to a fixed input size of 224x224 pixels to match the requirements of the EfficientNetB0 architecture. Image augmentation techniques such as rotation, zoom, and horizontal flipping were applied during training to improve generalization and reduce overfitting. Additionally, empty or unused folders were programmatically removed to ensure a clean training pipeline.

Label inconsistencies were addressed, and final class mappings were verified for consistency across all datasets. The dataset was normalized by rescaling pixel values to the $[0, 1]$ range.

4. Modeling

We employed the EfficientNetB0 convolutional neural network, a state-of-the-art architecture pretrained on ImageNet, and fine-tuned it for our classification task. Initially, the base layers of the model were frozen, and custom classification layers were added with dropout and batch normalization to prevent overfitting and enhance convergence.

After early epochs, the base model was unfrozen and fine-tuned using a smaller learning rate. This two-phase training approach significantly boosted performance, especially in later epochs. A combination of callbacks—including early stopping, learning rate reduction on plateau, and model checkpointing—ensured that training was efficient and optimal weights were preserved.

5. Evaluation

The model was evaluated using accuracy and categorical cross-entropy loss on the validation set. The onion model demonstrated significant improvements over time,

reaching a validation accuracy of over 92% after fine-tuning. The loss curve also confirmed convergence with occasional overfitting, mitigated through regularization and early stopping.

Performance was visualized using accuracy/loss curves and confirmed via single-image predictions. Test results showed strong alignment between actual and predicted labels, suggesting the model has learned meaningful features for disease identification.

6. Deployment

While full deployment is outside the current scope, the model has been saved in .keras format for future inference. A simple prediction pipeline has been implemented to allow users to input a new image and receive a predicted class label. This setup could be extended into a web or mobile app for field use, enabling farmers to take a photo and get instant feedback on crop health.

7. Conclusion and Next Steps

This CRISP-DM-based process successfully produced a high-performing image classification model for plant disease diagnosis in onions, maize, and tomatoes. Moving forward, the model can be improved by:

- Incorporating more diverse image data
- Expanding to other crops
- Evaluating model performance in real-world agricultural settings.

These enhancements will bring the project closer to real-world agricultural impact, supporting smarter and more responsive crop management.